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**Course Handout (Part II)**

Dated: 03/08/2026

In addition to Part-I (General Handout for all courses appended to the timetable) this portion gives further specific details regarding the course.

**Course No.** : INSTR F312  
**Course Title** : Transducers and Measurement Systems  
**Instructor-in-charge** : K. K. GUPTA ([kgupta@pilani.bits-pilani.ac.in](mailto:kgupta@pilani.bits-pilani.ac.in))  
**Lecture Instructors** : -----  
**Tutorial Instructors** : Dr. Sujan Yenuganti

**1. Course Description:**

INSTR F312 Transducers and Measurement Systems

Importance and types of measurement, generalized measurement system, functional elements, static and dynamic characteristics, primary sensing elements, modulating transducers, self generating transducers, inverse transducers, fiber optic transducers, measurement techniques for motion, seismic, pressure, flow, temperature, level, humidity, pH, viscosity; charge, isolation amplifier, MEMS based transducers;

**2. Scope and Objective of the course:** The objective of this course is to have a detailed study of measurement systems and to understand their application in the areas of monitoring, control and experimental engineering analysis.

**3. Learning Outcomes:**

- Analyze the functionalities and operations of sensors and measuring instruments techniques.
- Define static and dynamic performance metrics of sensor and instrumentation systems.
- Understand Working Principles of different sensing elements, such as Resistive sensing, Capacitive sensing, Inductive sensing, Electromagnetic sensing, Thermoelectric sensing, Hall effect and many other elements.
- Know how to select measurement devices for physical quantities like temperature, pressure, motion, and flow etc.

**4. Text Books and Reference Books:**

T1: Theory and Applications of Measurement Science, Surekha Bhanot, EDD Notes, Volume 1 and 2.

T2: Measurement Systems, E. O. Doebelin, Tata McGraw Hill Publication.

R1: Principles of Measurement Systems, John P. Bentley, Pearson India Education

R2: Doebelin's Measurement Systems, Ernest Doebelin, Dhanesh Manik,

R3: Instrumentation Measurement and Analysis, Nakra, Chaudhry

R4: Microsystem Design, Stephen D. Senturia.





## 5. Methods of conduct of the Course:

The course will be delivered by explaining theory through PowerPoint presentations and Board. The course also has tutorial component for solving numerical problems.

## 6. Course Plan:

| Module No.                                                 | Lec | Coverage                                                                                                                                                                                              | Ref                          | Outcome                                                                                      |
|------------------------------------------------------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|----------------------------------------------------------------------------------------------|
| 1. General Measurement System                              | 1   | Introduction to measurement systems, Structure of measurement systems, Examples of measurement systems                                                                                                | T1(1),<br>R1(1)<br>R2(1,2)   | Students will be able to understand the essence of measurement systems.                      |
| 2. Static Characteristics of Measurement System Elements   | 3   | Systematic characteristics, Generalized model of a system, Statistical characteristics, Identification of static characteristics – calibration                                                        | T1(2,3),<br>R1(2)<br>R2(2,3) | They will be able to identify and analyze the static characteristics of measurement systems. |
| 3. The Accuracy of Measurement Systems in the Steady State | 2   | Measurement error of a system of ideal elements, The error probability density function of a system of non-ideal elements, Error reduction techniques                                                 | R1(3)                        | They will be able to analyze the accuracy of measurement systems.                            |
| 4. Dynamic Characteristics of Measurement Systems          | 3   | Transfer function for typical system elements, Identification of the dynamics of an element, Dynamic errors in measurement systems, Techniques for dynamic compensation                               | T1(3),<br>R1(4)<br>R2(3)     | They will be able to analyze the dynamic characteristics of measurement systems.             |
| 5. Loading Effects and Two-port Networks                   | 3   | Effect of loading, Two-port networks                                                                                                                                                                  | R1(5)                        | They will be able to analyze the effect of loading, and model two port networks.             |
| 6. Signals and Noise in Measurement Systems                | 2   | Statistical representation of random signals, Effects of noise and interference on measurement circuits, Noise sources and coupling mechanisms, Methods of reducing effects of noise and interference | R1(6)                        | They will be able to understand the effect of noise in signals in measurement systems.       |





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|--------------------------------------------------|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|---------------------------------------------------------------------------------------------------|
| 7. Sensing Elements                              | 4 | Resistive sensing elements, Capacitive sensing elements, Inductive sensing elements, Electromagnetic sensing elements, Thermoelectric sensing elements, Elastic sensing elements, Piezoelectric sensing elements, Piezoresistive sensing elements, Electrochemical sensing elements, Hall effect sensors, Photoelectric Transducers | T1(4,5)<br>T1(6,7,8)<br>R1(8)<br>R2(4) | They will be able to identify sensing elements required in measurement systems.                   |
| 8. Signal Conditioning Elements                  | 1 | Brief about deflection bridges, Amplifiers, Op-Amp circuits, Instrumentation amplifier                                                                                                                                                                                                                                              | T1(11,12)<br>R1(9)<br>R2(9)            | They will be able to use various signal conditioning circuits to interface sensors.               |
| 9. Flow Measurement Systems                      | 4 | Essential fundamentals of fluid mechanics, Measurement of fluid velocity, volume flow rate, mass flow rate                                                                                                                                                                                                                          | T1(19)<br>R1(12)<br>R2(7)<br>R3(13)    | They will be able to analyze various flow measurement systems.                                    |
| 10. Pressure Measurement Systems                 | 3 | Fundamentals of pressure measurement, various types of pressure measurement systems                                                                                                                                                                                                                                                 | T1(18)<br>R2(6)<br>R3(11)              | They will be able to analyze various pressure measurement systems.                                |
| 11. Ultrasonic Measurement Systems               | 3 | Fundamentals of ultrasonic transmission, Ultrasonic sensors and measurement systems                                                                                                                                                                                                                                                 | R1(16)                                 | They will be able to understand the basics of ultrasonic measurement systems.                     |
| 12. Heat Transfer Effects in Measurement Systems | 2 | Basics of heat sensing, Dynamic characteristics of thermal sensors, Measurement systems based on heat sensing.                                                                                                                                                                                                                      | R1(14)                                 | They will be able to understand the fundamentals of heat transfer effects in measurement systems. |
| 13. Optical Measurement Systems                  | 1 | Optical sensors and fundamentals of optical measurement systems, Optical fiber                                                                                                                                                                                                                                                      | R1(15)                                 | They will be able to understand the fundamentals of optical measurement systems.                  |
| 14. Motion and Vibration Measurement Systems     | 2 | Fundamentals of vibration, Measurement of various types of motion, Vibration measurement systems                                                                                                                                                                                                                                    | R2(4)<br>R3(7)                         | They will be able to analyze various motion and vibration measurement systems.                    |





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|--------------------------------------------------------------|---|-------------------------------------------------------------------------------------|-----------------|--------------------------------------------------------------------------------|
| 15. Humidity, pH, Viscosity Measurement                      | 2 | Essential fundamentals of humidity, pH and viscosity, and their measurement systems | T1(22)<br>R2(7) | They will be able to analyze humidity, pH and viscosity measurement.           |
| 16. MEMS                                                     | 1 | Fundamentals of MEMS, SOC                                                           | T1(10)<br>R4(1) | They will be able to understand the contemporary MEMS technology and SOC.      |
| 17. Reliability, Choice and Economics of Measurement Systems | 2 | Reliability and choice of measurement systems, Total lifetime operating cost        | R1(7)           | They will be able to understand the reliability factor of measurement systems. |

**NB:** The treatment and the depth of coverage of certain topics may vary slightly from what is given in the textbook and reference books. Tests will be based on what will be covered in the lecture and the tutorial hours.

#### 7. Evaluation Scheme:

| Component                 | Duration  | Weightage (%) | Date & Time  | Nature of component (Close Book/ Open Book) |
|---------------------------|-----------|---------------|--------------|---------------------------------------------|
| Mid-Semester Test         | 90 Min.   | 30            | As per AUGSD | Closed Book                                 |
| Quiz                      | 10-15 Min | 15            | Continuous   | Closed Book                                 |
| Assignment/Project        | -         | 10            |              | Open Book                                   |
| Comprehensive Examination | 180 Min.  | 45            | As per AUGSD | Open/Closed Book                            |

**8. Chamber Consultation Hour:** To be announced during the class.

**9. Course Notices:** All notices of this course will be displayed on NALANDA; Make-up will be given on extremely genuine grounds only. An application should be made in advance to request the make-up examination.

Instructor-in-Charge  
**INSTR F312**

